

CE 310

Excel Tutorial — Getting Started with Data

Week 1 · Session 2 · Hands-on Walkthrough

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today's Plan — Excel Walkthrough, Then You're on Your Own

By the end of this walkthrough you will have:

1. The dataset open and verified in Excel
2. An understanding of cell reference notation
3. Five statistical formulas producing the same values from Monday's lecture
4. A `Month` helper column built and filled for all 8,760 rows
5. `COUNTIF` and `AVERAGEIF` working

Then you open the in-class exercise and work independently.

Everything we do today is also on the last slide of Monday's lecture as a reference card.

 **Keep this file.** Several slides show a picture of the spreadsheet as it should look at that point. If a number comes out wrong later in the week, find the matching picture and compare it against your screen — that is faster than re-reading the formula.

Step 1 — Open the File

File name: `Week1_Tucson_Weather_2023.csv`

Option A — Simplest:

File → Open → navigate to the file → Open

Excel detects it is a CSV and loads it automatically.

Option B — More control:

Data tab → From Text/CSV → select the file → Load

Use Option B if Option A gives garbled columns.

What you should see after opening:

- Row 1: headers — `DateTime`, `Temp_F`, `Wind_Speed_mph`, `Relative_Humidity_pct`
- Rows 2–8761: one row per hour of 2023
- **8,760 data rows** = 365 days × 24 hours

⚠ Save immediately as an **.xlsx** file (File → Save As). CSV files do not save formulas.

Step 1 — Check Before You Go On

What you should see after opening the file

	A	B	C	D
1	DateTime	Temp_F	Wind_Speed_mph	Relative_Humidity_pct
2	2023-01-01 00:00	69.1	3.4	37.5
3	2023-01-01 01:00	63	0	48
4	2023-01-01 02:00	62.1	4.7	49.6
⋮				
8760	2023-12-31 22:00	63	4.7	26.6
8761	2023-12-31 23:00	61	4.7	32

← Row 1 is the header — column names, not data

Rows 2-8761
8,760 hours = 365 × 24

Ctrl + End jumps to the last cell. If it does not land on row 8761, the file did not load cleanly.

If your columns are all crammed into column A, the CSV was read with the wrong delimiter — close it and reopen with **Option B**.

Step 2 — Verify and Set Up

Before writing any formula, spend 60 seconds verifying the file loaded correctly.

Check the row count:

Press **Ctrl + End** — the cursor jumps to the last data cell. The row number should be **8761** (row 1 is the header).

Freeze the header row:

View → Freeze Panes → Freeze Top Row

Now row 1 stays visible no matter how far you scroll down.

Check column D:

Scroll right — column D should be **Relative_Humidity_pct** with values roughly between 10% and 90%.

Label a results area:

Click on cell **F1** and type **Results** — this is where you will put your formulas so they don't overwrite the data.

Step 3 – Cell Reference Notation

Every cell has an address: **column letter + row number**.

Reference	What it points to
B1	The header cell — says "Temp_F"
B2	The first temperature value (hour 1 of 2023)
B8761	The last temperature value (hour 8,760)
B2:B8761	Explicit range — all 8,760 temperature readings, no header
B:B	Whole column — safe in COUNTIF (ignores text), avoid in AVERAGE

Step 3 — Getting the Range Right

One column, three ways to write the range

`=AVERAGE(B2:B8761)`

	B
1	Temp_F
2	69.1
3	63
⋮	
8761	61

✓ **8,760 numbers**

What every formula in this course should use.

`=AVERAGE(B1:B8761)`

	B
1	Temp_F
2	69.1
3	63
⋮	
8761	61

✗ **Includes the header**

Starts at row 1, so the text "Temp_F" is inside the range.

`=COUNTIF(B:B,">95")`

	B
1	Temp_F
2	69.1
3	63
⋮	
8761	61

⚠ **The whole column**

Safe in COUNTIF, which skips text. Avoid it in AVERAGE.

Almost every wrong answer in Week 1 is a range that starts on row 1 instead of row 2.

Step 4 — Your First Formula

Type along with me

Click on cell **G2**. Type exactly:

```
=AVERAGE(B2:B8761)
```

Press **Enter**.

You should see approximately **72** (the exact 2023 value is **72.0°F**). If your result is between 71°F and 73°F, you are correct. That is the annual average temperature discussed on Monday.

Does your result match? Raise your hand if you got something outside the 71–73°F range.

⚠ A result of **0** or **#DIV/0!** almost always means the range started at row 1 — the header text is not a number. Use **B2:B8761**.

Step 4 — Reading the Formula Bar

The formula lives in G2 — the numbers live in B

G2	fx =AVERAGE(B2:B8761)						
	A	B	C	D	E	F	G
1	DateTime	Temp_F	Wind...	Rel_Hum...		Results	
2	2023-01-01 00:00	69.1					72.0
3	2023-01-01 01:00	63					
⋮							
8761		61					

Type in G2, read from B2:B8761. Keeping results away from the data is why Step 2 labelled F1 “Results”.

The formula bar at the top always shows what is really in the selected cell — the grid shows only the answer.

Step 5 — Eight Core Formulas (Type Along)

In the cells below G2, enter each formula. The results should match your lecture notes exactly.

Cell	Formula	Expected result	Tolerance
G2	=AVERAGE(B2:B8761)	≈ 72.0 °F	± 0.5 °F
G3	=MEDIAN(B2:B8761)	≈ 73.0 °F	± 0.5 °F
G4	=MAX(B2:B8761)	≈ 112 °F	exact
G5	=MIN(B2:B8761)	≈ 28 °F	exact
G6	=STDEV(B2:B8761)	≈ 18.0 °F	± 0.5 °F
G7	=LARGE(B2:B8761, 88)	≈ 108 °F	exact
G8	=QUARTILE(B2:B8761, 1)	≈ 58 °F	± 1 °F
G9	=QUARTILE(B2:B8761, 3)	≈ 86 °F	± 1 °F
G10	=QUARTILE(B2:B8761, 3) - QUARTILE(B2:B8761, 1)	≈ 28 °F	± 2 °F

Step 5 — What These Numbers Mean

If your result is outside the tolerance, the most common cause is including row 1 (the header) in your range — change `B2:B8761` and check again.

Why 88? $1\% \times 8,760 = 87.6 \rightarrow$ round up $\rightarrow$ 88th-largest = ASHRAE 1% design temperature.

QUARTILE arguments: 1 = Q1 (25th percentile), 3 = Q3 (75th percentile). The difference $Q3 - Q1$ is the **IQR** — the range covered by the middle 50% of the data.

Step 6 — Build the MONTH Helper Column

Many of the exercise questions need to know which month each row belongs to. Excel can extract this from the `DateTime` column automatically.

Do this now:

1. Click cell **E1** → type `Month` (the column label)
2. Click cell **E2** → type `=MONTH(A2)` → press Enter
3. You should see **1** (January — the first hour of 2023)

`=MONTH(date)` extracts the month number: January = 1, July = 7, December = 12.

Why column E? Columns A–D are raw data; E is the first free column. Keep helper columns beside the data so your ranges stay tidy.

Step 6 — What You Should See

Step 6 — one helper column, built from the timestamp

E2	fx =MONTH(A2)				
	A	B	C	D	E
1	DateTime	Temp_F	Wind...	Rel_Hum...	Month
2	2023-01-01 00:00	69.1			1
3	2023-01-01 01:00	63			
⋮					
8761					

reads this cell...

...returns 1

January = 1, July = 7, December = 12. Column E is the first free column, so the raw data in A-D stays untouched.

One formula, one cell — the fill in Step 7 turns it into all 8,760.

Step 7 — Filling Down to All 8,760 Rows

You have `=MONTH(A2)` in E2. You need it in E3 through E8761.

Method 1 — Double-click the fill handle

(fastest)

Select E2, move the cursor to the small square at its bottom-right corner, and double-click. The next slide shows exactly what to look for.

Method 2 — Keyboard shortcut

Select E2, hold **Shift**, click E8761, then press **Ctrl + D** (fill down).

💡 When you fill down, Excel adjusts the row number automatically: E3 gets `=MONTH(A3)`, E4 gets `=MONTH(A4)`, and so on. This is called a **relative reference** — the formula adapts to each new row.

⚠️ Check the bottom before you move on. Press **Ctrl + End** — the last row must be **8761**. A fill that stopped early leaves every formula reading this column quietly wrong, with nothing on screen to tell you.

Step 7 — The Fill Handle, Step by Step

Filling one formula down 8,760 rows

1

	E
2	1
3	
4	
⋮	
8761	

Select E2. The small square at its bottom-right is the fill handle.

2

	E
2	1
3	
4	
⋮	
8761	

Move onto the handle — the pointer becomes a thin black cross.

3

	E
2	1
3	1
4	1
⋮	
8761	12

Double-click. Excel fills to the last row with data beside it.

Check it worked: Ctrl + End lands on E8761, which should read 12 — December.
Keyboard alternative: select E2, Shift-click E8761, then Ctrl + D.

Step 8 – COUNTIF: Count by Condition

`=COUNTIF(range, criteria)` counts rows where one condition is true.

The **criteria** goes in quotes whenever it uses a comparison operator:

Formula	What it counts	Result
<code>=COUNTIF(B2:B8761, ">95")</code>	Hours above 95 °F	≈ 900
<code>=COUNTIF(B2:B8761, ">100")</code>	Hours above 100 °F	≈ 429
<code>=COUNTIF(E2:E8761, 7)</code>	Hours in July	744

⚠ Most common mistake — forgetting the quotes:

`=COUNTIF(B2:B8761, >95)` → Excel shows a **#VALUE!** error

`=COUNTIF(B2:B8761, ">95")` → returns ≈ 900 ✓

Step 9 — COUNTIFS: Count with Multiple Conditions

`=COUNTIFS(range1, criteria1, range2, criteria2, ...)` counts rows where **all** conditions are true at the same time. Add as many range/criteria pairs as you need.

Engineering question: How many hours had both warm temperatures and high humidity — Tucson's monsoon comfort burden?

Formula	What it counts	Result
<code>=COUNTIFS(B2:B8761,">80",D2:D8761,">50")</code>	Temp > 80 °F and RH > 50%	≈ 144
<code>=COUNTIFS(B2:B8761,">75",D2:D8761,">60")</code>	Temp > 75 °F and RH > 60%	≈ 116

Setup: column B = `Temp_F`, column D = `Relative_Humidity_pct`

Step 9 — Which Rows Get Counted

COUNTIFS — both conditions, on the same row

```
=COUNTIFS(B2:B8761, ">80", D2:D8761, ">50")
```

	Temp > 80 °F		RH > 50 %	
	B	C	D	
2	82.4	3.4	55.0	82.4 and 55.0 — counted
3	78.1	0.0	62.0	78.1 fails the temperature test
4	85.0	4.7	48.0	48.0 fails the humidity test
5	91.2	2.1	57.0	91.2 and 57.0 — counted
⋮				
8761				

Both ranges must cover the same rows. D2:D8760 against B2:B8761 is one row short, and Excel refuses the whole formula.

Step 10 — AVERAGEIF: Average by Condition

`=AVERAGEIF(criteria_range, criteria, average_range)` computes the mean for a specific subset of rows.

Three arguments — in order:

1. **Where to look** for the condition → column E (month numbers)
2. **What to match** → 7 for July, 1 for January
3. **What to average** when the condition is met → column B (temperature)

Formula	What it computes	Result
<code>=AVERAGEIF(E2:E8761, 7, B2:B8761)</code>	July average temperature	≈ 94 °F
<code>=AVERAGEIF(E2:E8761, 1, B2:B8761)</code>	January average temperature	≈ 51 °F

Step 10 — Which Column Is Which

AVERAGEIF — three arguments, three columns

^①
=AVERAGEIF(^②E2:E8761, ^③7, B2:B8761)

③ what to average

① where to look

	B	C	D	E
2	51.0			1
3	93.5			7
4	96.2			7
5	49.8			1
⋮				
8761				

② rows where the month is 7
→ average 93.5 and 96.2, not all 8,760

July average = 94.3 °F. The criteria column and the averaged column must cover the same rows.

Step 11 — STDEV(IF): Standard Deviation by Condition

Excel has no `STDEVIF` function. Instead, combine `STDEV` and `IF` as an **array formula**.

 **Enter this formula for July's standard deviation:**

Click a blank cell → type `=STDEV(IF(E2:E8761=7, B2:B8761))`

Press **Ctrl + Shift + Enter** (not just Enter)

Excel wraps the formula in curly braces: `{=STDEV(IF(E2:E8761=7, B2:B8761))}`

Expected result: ≈ 9.4 °F

If you have Excel 365 or Excel 2021, you can press Enter normally:

```
=STDEV(FILTER(B2:B8761, E2:E8761=7))
```

Step 11 — STDEV(IF): Results & Common Mistake

Now try January (month = 1):

Formula	Month	σ result
<code>{=STDEV(IF(E2:E8761=7, B2:B8761))}</code>	July	$\approx 9.4\text{ }^{\circ}\text{F}$
<code>{=STDEV(IF(E2:E8761=1, B2:B8761))}</code>	January	$\approx 10.3\text{ }^{\circ}\text{F}$

These are B3 and B4 in this week's exercise — you have just computed them.

Note how close they are: January's hourly temperatures are only about 9% more spread out than July's. That is *not* what the Quick Vote found, and the difference is the point. The vote asked which month varies more **day to day**, and the answer there was January by 2.4 $\times$. Same two months, two different questions — because σ of every hourly reading and σ of the daily averages measure different things. July's hours swing widely between night and afternoon but every day repeats that same cycle; January's days genuinely differ from one another. Averaging to daily values first strips out the within-day swing and leaves only the day-to-day part.

Step 12 — Charts Plot Cells, Not Pictures

Both charts this week want a **flat reference line** at a value you computed. Excel has no button for that, and looking for one is the wrong instinct.

An Excel chart draws **cells**. Anything you want on a chart has to exist as cells first — including a straight line.

A reference line is a **column holding the same number over and over**:

Month	Avg Temp (°F)	Annual avg (°F)
1 — January	55.6	72.0
2 — February	58.9	72.0
...	...	72.0

Twelve rows, one value, repeated. That column *is* the line.

Step 13 — Building Chart 1

On the **Work** sheet, with your twelve monthly averages already in **B7:B18**:

1. **C6** — type a header: **Annual avg (°F)**
2. **C7** — type your A1 answer (≈ 72.02)
3. Copy **C7** , select **C8:C18** , paste — twelve identical values
4. Select **A6:C18** , headers included
5. **Insert** → **Charts** → **Line** (or **Column** — either is accepted)

Excel draws **two series**, because your selection had two value columns. The flat one is your reference line.

💡 Change a value in **C7:C18** and watch the line move. It is tied to your number — that is the whole point.

Step 13 — Finishing the Chart

Every chart in this course is scored on the same four things. All of them live in one menu: **Chart Design** → **Add Chart Element**.

Element	Where	What to put
Chart Title	Add Chart Element → Chart Title	Monthly Average Temperature – Tucson, AZ 2023
Axis Titles	Add Chart Element → Axis Titles	Month (1–12) and Average Temperature (°F)
Legend	Add Chart Element → Legend	Usually added for you once there are two series
Data Labels	Add Chart Element → Data Labels	Only when a week asks for them

Then right-click the flat series → **Format Data Series** → dashed line. A reference is not data and should not look like data.

 **Do not draw the line with *Insert* → *Shapes*.** It looks identical and is attached to nothing: correct your A1 later and the drawn line stays where it was, silently wrong.

Step 13 — Units Go in the Label

Every axis title in this course carries its units. `Temperature` is not enough;
`Average Temperature (°F)` is.

This is not formatting fussiness. A number without a unit is not an engineering result — 72 could be °F, °C, or a count of hours, and only the label says which. Charts are read by people who did not build them.

💡 The same rule applies to your memo. Every figure you cite carries its unit.

Something's Wrong — Start Here

Four things that go wrong — and what they look like

`=COUNTIF(B2:B8761, >95)`

	H
2	#VALUE!

The criteria has an operator but no quotes.

fix: `=COUNTIF(B2:B8761, ">95")`

`=COUNTIFS(B2:B8761, ">80", D2:D8760, ">50")`

	H
2	#VALUE!

The two ranges are different lengths — 8760 vs 8761.

fix: `..., D2:D8761, ">50")`

`=STDEV(IF(E2:E8761=7, B2:B8761))`

	H
2	#VALUE!

Entered with Enter. This one needs Ctrl + Shift + Enter.

fix: `{=STDEV(IF(E2:E8761=7, B2:B8761))}`

column A, after opening the file

	A
2	#####

Not an error — the column is too narrow to print the date.

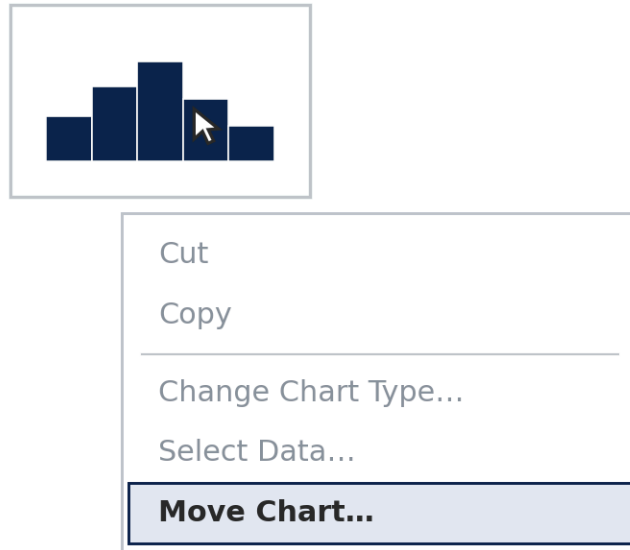
fix: double-click the A|B divider

The Step Every In-Class Exercise Ends With

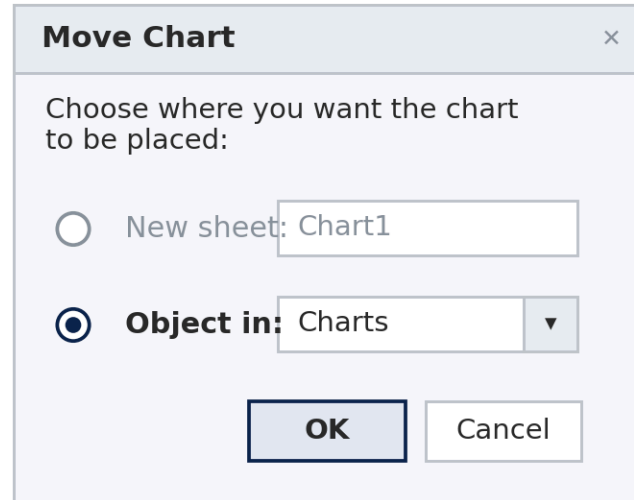
Getting a finished chart into the box it is graded in

Every week. Build the chart on the sheet its data lives on, then move it onto the Charts sheet.

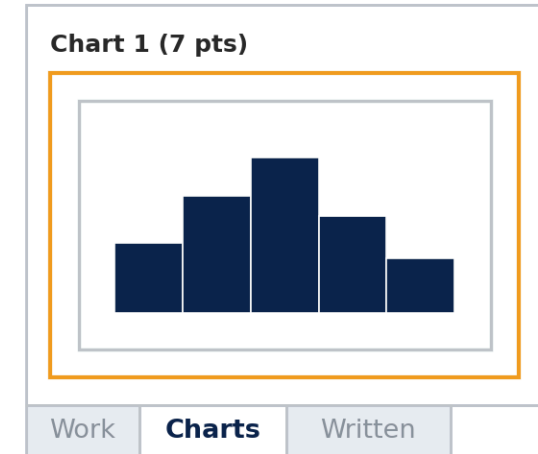
1 Right-click the chart



2 Choose "Object in"



3 It lands in the box



✗ New sheet — makes a separate sheet called Chart1. The box on Charts stays empty and the points are lost.

✓ Object in → Charts — then drag the chart inside the outlined box.

⚠ Leave it a live chart — a pasted picture of one scores zero.

Quick Reference — Statistics & Helper Column

Statistics (column B = Temp_F):

Task	Formula	Result
Mean	=AVERAGE(B2:B8761)	≈ 72 °F
Median	=MEDIAN(B2:B8761)	≈ 73 °F
Max / Min	=MAX / =MIN	112 / 28 °F
Std deviation	=STDEV(B2:B8761)	≈ 18 °F
Q1 / Q3	=QUARTILE(B2:B8761, 1/3)	58 / 86 °F
IQR	=QUARTILE(..., 3) - QUARTILE(..., 1)	≈ 28 °F
ASHRAE 1% design temp	=LARGE(B2:B8761, 88)	≈ 108 °F
Extract month	=MONTH(A2) → fill down	1–12

Quick Reference — Conditional Functions

Conditional functions (column E = Month):

Task	Formula
Count by 1 condition	<code>=COUNTIF(B2:B8761, ">95")</code>
Count by 2+ conditions	<code>=COUNTIFS(B2:B8761, ">80", D2:D8761, ">50")</code>
Average by condition	<code>=AVERAGEIF(E2:E8761, 7, B2:B8761)</code>
Std dev by condition	<code>{=STDEV(IF(E2:E8761=7, B2:B8761))}</code>

Quick Reference — Navigation

Task	Shortcut
Open CSV	File → Open (save as .xlsx)
Freeze header	View → Freeze Panes → Top Row
Jump to last row	Ctrl + End
Fill down	Double-click fill handle or Ctrl+D

Before You Submit

⚠ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

1. **File format** — **.xlsx only**. Save → Save As → Excel Workbook (*.xlsx). Files saved as **.xls**, **.csv**, or **.pdf** cannot be graded.
2. **Sheet name** — **Answers**. Do not rename or delete this sheet.
3. **Column layout** — **do not change**. Answers go in Column E. Do not insert, delete, or move columns.
4. **File name** — **CE310_W1_<NetID>.xlsx**, e.g., **CE310_W1_abc123.xlsx**. Fill in the **NetID** cell at the top of the **Answers** sheet as well — that cell is what matches your work to your student record, and a 5-point deduction applies if it is blank. (The weekly assignment is a separate file, named **CE310_W1_Assignment_<NetID>.xlsx**.)
5. **Upload** the **.xlsx** file to **D2L** → **Assignments** by **9:00 AM, Wed Sep 2**.

You're Ready — Open the In-Class Exercise

Wk01.03_In_Class_Exercise.pdf

Your file should now have:

- Column E filled with month numbers (1–12)
- A results area in column G with AVERAGE, MEDIAN, MAX, MIN, STDEV, QUARTILE, IQR, LARGE
- Spot-checks using COUNTIF, COUNTIFS, AVERAGEIF, and STDEV(IF)

The exercise picks up from here. **Section A** asks for the same formulas you just typed — you are already partway done.

Work independently. Raise your hand if you get stuck on a formula or a result that doesn't match the expected value.